

Enterprise Resource Planning (ERP) System for Small Business - Detailed Requirement Analysis

The ERP system is designed to manage and automate various business processes across finance, HR, inventory, and sales. The goal is to create a centralized platform that enables efficient business operations for small businesses, streamlining tasks and improving decision-making through real-time reporting and analytics.

Key Functional Areas and Entities

1. Employee

- Stores employee information such as name, contact details, position, department, salary, and attendance.
- Related to: Department, Leave, Salary, Role.

2. Department

- Manages different business units like HR, Sales, Finance, and Inventory. Each department will have a set of employees and roles.
- Related to: Employee, Role.

3. Product

- Contains product information such as name, SKU, price, quantity, and supplier details.
- Related to: Supplier, Inventory, SalesOrder, PurchaseOrder.

4. Supplier

- Manages supplier details, including company name, contact information, and products supplied.
- Related to: Product, PurchaseOrder, Payment.

5. Customer

- Manages customer data such as name, contact, order history, and payment status.
- Related to: SalesOrder, Invoice, Payment, Shipment.

6. SalesOrder

- Represents customer orders, including product details, order status, and customer information.
- Related to: Customer, Product, Inventory, Invoice, Payment, Shipment.

7. PurchaseOrder

- Manages purchase requests to suppliers, including products, quantities, and delivery dates.
- Related to: Supplier, Product, Inventory, Payment.

8. Inventory

- Tracks the stock levels of products, incoming supplies, and outgoing sales.
- Related to: Product, PurchaseOrder, SalesOrder.

9. Invoice

- Generates billing for sales or purchases, including details of products, amounts, and tax calculations.
- Related to: SalesOrder, PurchaseOrder, Payment, Customer, Supplier.

10. Payment

- Manages financial transactions, including payments to suppliers and customer payments for orders.
- Related to: Invoice, Supplier, Customer, SalesOrder, PurchaseOrder.

11. Leave

- Tracks employee leave requests and approvals, including sick leave, vacation, and other types of leave.
- Related to: Employee, Department.

12. Salary

- Manages payroll data for employees, including salary calculation, bonuses, deductions, and tax.
- Related to: Employee, Payment, Department.

13. Shipment

- Handles the shipment of products to customers or the receipt of products from suppliers.
- Related to: SalesOrder, PurchaseOrder, Inventory, Customer, Supplier.

14. Role

- Defines user roles within the system, such as Admin, HR Manager, Salesperson, Inventory Manager, etc., with associated permissions.
- Related to: Employee, Department, User.

15. User

- Manages the authentication and authorization of system users, with login credentials and role-based access.
- Related to: Role, Employee.

Modules and Functional Analysis

1. HR Management Module

The HR module is responsible for managing employees, departments, payroll, and attendance. This module integrates seamlessly with other parts of the system (e.g., payroll integration with finance).

Functionalities:

- **Employee Management:**
 - Add/edit/delete employee records.
 - Assign roles, departments, and manage employment status.
 - Manage personal information and job details.
- **Department Management:**
 - Create and manage departments.
 - Assign employees to departments.
 - Track department performance.

- **Attendance and Leave Management:**
 - Track employee attendance and absences.
 - Process leave requests, approvals, and rejections.
 - Automate attendance-based salary deductions.
 - **Payroll Processing:**
 - Calculate employee salaries based on attendance, leave, and role-based wages.
 - Manage bonuses, deductions, and allowances.
 - Generate salary slips and payment invoices.
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2. Inventory Management Module

The Inventory module manages product stocks, supplier relationships, and ensures that stock levels are maintained for smooth sales operations. It ensures product availability and reduces excess stock.

Functionalities:

- **Product Management:**
 - Add/edit/delete products.
 - Assign SKUs, price, and stock levels.
 - Track product categories and suppliers.
 - **Supplier Relationship Management:**
 - Manage supplier information and product associations.
 - Track purchase orders and supplier performance.
 - **Stock Control:**
 - Automatically update stock levels based on purchase and sales orders.
 - Generate low-stock alerts.
 - Manage inventory audits and stock adjustment.
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3. Finance Management Module

This module handles all financial transactions, including billing, invoicing, and payments. It helps maintain financial records, ensuring accuracy and compliance.

Functionalities:

- **Invoicing:**
 - Generate invoices for sales orders, including product details, tax calculations, and customer details.
 - Track paid and unpaid invoices for suppliers and customers.
- **Billing and Payments:**
 - Manage customer payments for sales orders.
 - Track and process supplier payments for purchase orders.

- Integrate with external payment gateways for easy transactions.
 - **Financial Reporting:**
 - Generate financial statements such as balance sheets, cash flow, and expense reports.
 - Track income and expenses by department, supplier, or product.
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4. Sales Management Module

The sales module focuses on managing customer interactions, processing sales orders, and ensuring timely order fulfillment and invoicing.

Functionalities:

- **Sales Order Processing:**
 - Create, edit, and manage customer sales orders.
 - Track order statuses (e.g., pending, fulfilled, cancelled).
 - Automatically adjust inventory after sales completion.
 - **Customer Management:**
 - Manage customer profiles, order history, and contact information.
 - Track customer feedback and manage complaints.
 - **Order Fulfillment and Shipment:**
 - Track shipments for sales orders.
 - Generate shipment orders and delivery schedules.
 - Send notifications to customers upon order dispatch.
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5. Reporting Module

The reporting module aggregates data from all other modules and generates meaningful reports to aid in decision-making. Real-time insights into business operations help in improving efficiency and profitability.

Functionalities:

- **HR Reports:**
 - Employee performance reports (based on attendance, department, etc.).
 - Payroll reports and salary breakdowns.
 - Department-wise employee count and productivity reports.
- **Inventory Reports:**
 - Stock levels, reorder alerts, and slow-moving products.
 - Supplier performance reports (on-time deliveries, quality issues).
 - Purchase and sales report for inventory turnover analysis.
- **Finance Reports:**
 - Income, expense, and profit-loss reports.

- Unpaid invoices, accounts payable, and receivable reports.
 - Departmental financial performance tracking.
 - **Sales Reports:**
 - Sales order reports based on customers, products, and salespersons.
 - Order fulfillment reports and delivery time tracking.
 - Customer satisfaction and feedback analysis.
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System Design and Workflow

The ERP system will follow a **modular architecture** where each module is responsible for specific tasks but shares data with other modules to create a unified platform.

1. **User Login and Role Management:** Users can log in based on their roles (Admin, HR, Sales, Inventory Manager, etc.), which will control their access to specific features in the system.
2. **Centralized Database:** The system will maintain a centralized database where all data (employees, customers, products, orders, etc.) are stored, ensuring that each module has access to up-to-date information.
3. **Data Flow Example:**
 - A customer places an order → Sales Module creates a SalesOrder → Inventory Module checks stock availability → Finance Module generates an invoice → Shipment Module manages delivery → Finance Module tracks payment.
4. **Integration Points:**
 - HR and Payroll integrate with Finance for salary payments.
 - Sales Orders integrate with Inventory and Shipment for order fulfillment.
 - Purchase Orders integrate with Supplier and Inventory for stock replenishment.
 - Reporting pulls data from all modules to create real-time reports.